Building Representative Matched Samples With Multi-Valued Treatments in Large Observational Studies

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Chile's Challenges

- **Income Inequality:** Chile has the highest level of income inequality (measured by the Gini index) among the all OECD countries.
- **Education System:** A college degree is crucial in Chile. College admission and scholarship is largely determined (about 80%) by a single high-stakes exam called PSU, that is offered only once per year.
- **Earthquakes:** Chile experiences a major earthquake (magnitude $\geq$ 7.0) on average more than once every two years.

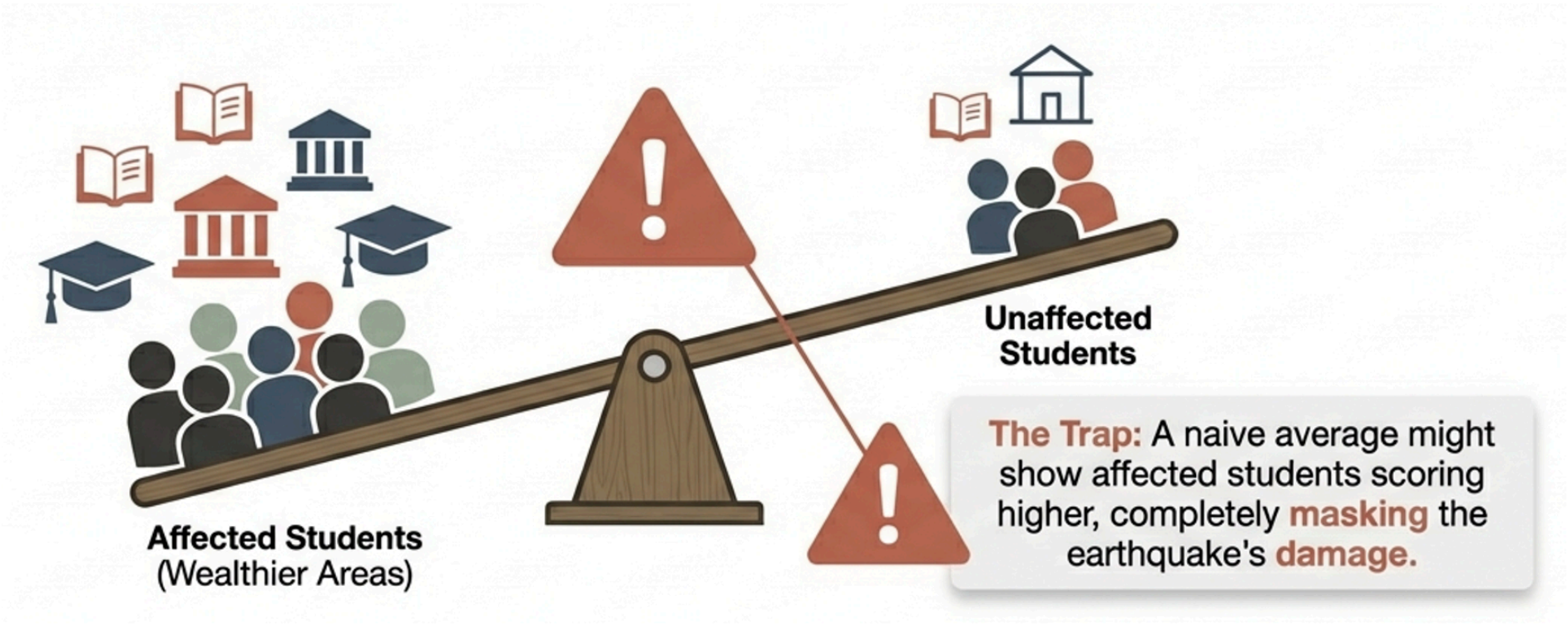
Research Objective

- This study examines the causal effect of the 2010 Chile earthquake, which resulted in more than 500 fatalities, on high school attendance and PSU examination scores.
- Can we just take averages of affected and unaffected students and see their difference?

The Illusion of Averages

- A major wildfire struck Los Angeles in January 2024.
- The damage was concentrated in Pacific Palisades and Altadena.
- Students in South Central Los Angeles and Compton were largely unaffected.
- Can differences in average SAT scores meaningfully capture the wildfires true impact?

The Illusion of Averages



The Confounding

- Earthquake-affected areas may differ systematically from unaffected areas (e.g., in wealth or school quality).
- These pre-existing differences may also influence test scores.
- When factors other than the treatment (the earthquake exposure) also affect the outcome, this problem is known as *confounding* in causal inference.

Matching in Causal Inference

- To make a credible causal claim, treatment and control groups must be comparable in all relevant characteristics.
- Ideally, we want to observe the same unit in parallel universes–with and without the treatment.
- In practice, we use *matching*: we pair each treated unit with a similar control unit based on observed characteristics.
- Once units are comparable, the difference in means becomes more meaningful as an estimate of the treatment effect.

Contribution of This Paper

- The dominant matching approaches rely on network flow methods that minimize covariate distance.
- Although mixed-integer programming (MIP) approaches exist, they have been less widely adopted, largely because NP-hard problems are often viewed as computationally infeasible in practice.
- This paper proposes a direct matching approach based on *cardinality* rather than distance minimization.
- They formulate the problem as a MIP and introduce reformulation techniques that substantially improve computational speed.

Methodology: Data and Setup

- Census data from Chile ($\sim$100,000 students).
- Treatment: earthquake exposure (3, 5, and 10 levels).
- Each student is described by 14 categorical covariates:
  - Gender
  - Ethnicity
  - Parental education
  - Household income (discretized)
  - etc.

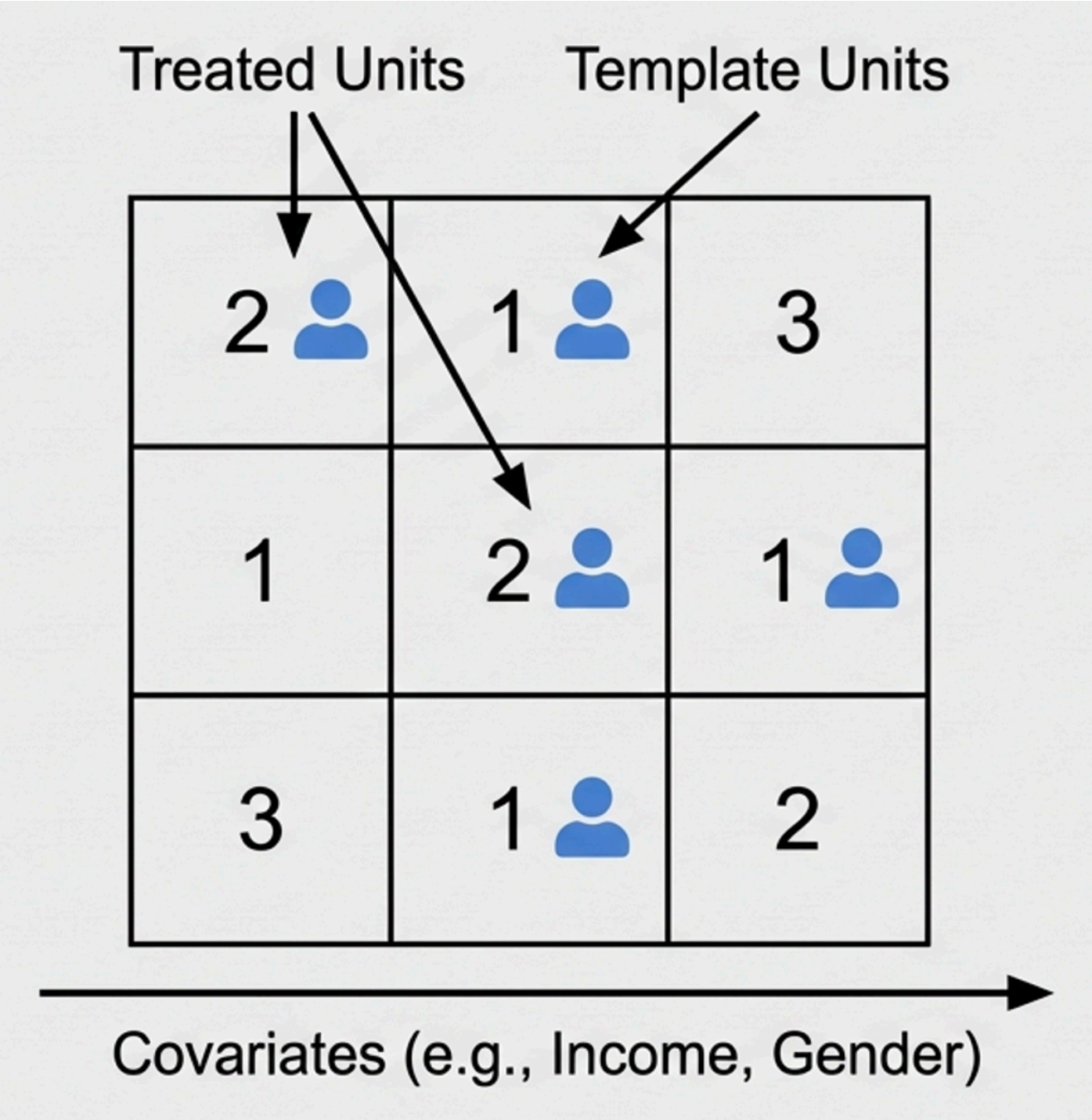
Multiple Treatments and the Template Dataset

- With two treatment levels, one group can serve as a reference.
- With multiple treatment levels, pairwise matching is cumbersome.
- Solution: construct a *template population*.
  - Hypothetical sample representing Chiles overall covariate distribution.
  - Each treatment level is matched separately to this template.
- By repeating randomized drawing of template population, we can do hypothesis testing and build confidence interval
- Fix one treatment level in this presnetation, for simplicity

Cardinality Matching via MIP

- Think of the data as a $P \times K$ grid:
  - Rows = covariates
  - Columns = categories
- The template determines how many units belong in each cell.
- We choose treated units to "fill" the grid so that the cell counts match the template as closely as possible.
- The objective: minimize total imbalance across all cells.

This approach is known as **cardinality matching**.


Treated Units
Template Units
2
1
3
1
2
1
3
1
2
Covariates (e.g., Income, Gender)

Cardinality Matching via Mixed-Integer Programming

1. **Exact Match:** Each template unit matches exactly one treated unit
2. **No Reuse:** Each treated unit can match at most one template unit

Notations

- We consider P categorical covariates, $\mathcal{P} = \{p_1, \ldots, p_P\}$
- For ease of exposition we assume that each has exactly K categories.
- Accordingly, each unit i is represented by a covariate vector, for example, $\mathbf{x}_i \in [K]^P$.

Notations

We haave two dataset:

1. Treatment Dataset ($\mathcal{L}$)
   - Units that received the specific treatment level under study
   - We denote this set as $\mathcal{L} = \{\ell_1, \ldots, \ell_L\}$
   - For example, a unit $\ell \in \mathcal{L}$ consists of an outcome y_ℓ, a treatment A_ℓ, and a covariate vector $\mathbf{x}_\ell$.
2. Template Dataset ($\mathcal{T}$):
   - A sample drawn from the population that represents the target covariate distribution
   - We denote this set as $\mathcal{T} = \{t_1, \ldots, t_T\}$
   - Since this dataset is generated for matching purposes, we only care about the covariate vector $\mathbf{x}_t$ for a given unit $t \in \mathcal{T}$.

Notations

- Problem Coefficients: For each covariate $p \in \{1, \ldots, P\}$ and category $k \in \{1, \ldots, K\}$, we define $N_{p,k}$ as The number of units in the $\mathcal{T}$ that belong to category k of covariate p. These represent our target counts for a representative sample.
- Decision Variables:The optimization model uses two types of variables:
  - $m_{t,\ell} \in \{0, 1\}$: A binary indicator that is 1 if template unit $t \in \mathcal{T}$ is matched to treated unit $\ell \in \mathcal{L}$, and 0 otherwise.
  - $v_{p,k} \geq 0$: A continuous variable representing the "cell count margin"the absolute difference between the number of matched treated units and the target count $N_{p,k}$.
- Objective: Minimize $\sum_{p,k} v_{p,k}$

Constraints

1. Count Balance: This constraint connects the matching decisions (m) to the discrepancy margins (v):

$$|\sum_{\ell \in \mathcal{L}_{p,k}} \left(\sum_{t \in \mathcal{T}} m_{t,\ell} \right) - N_{p,k}| \leq v_{p,k}$$

   where $\mathcal{L}_{p,k}$ is the subset of treated units possessing category k of covariate p.

2. Matching Requirements: These ensure a valid one-to-one mapping without replacement:
   1. Each treated unit is matched at most once: $\sum_{t \in \mathcal{T}} m_{t,\ell} \leq 1, \quad \forall \ell \in \mathcal{L}$
   2. Each template unit is matched exactly once: $\sum_{\ell \in \mathcal{L}} m_{t,\ell} = 1, \quad \forall t \in \mathcal{T}$

Optimization problem

Collecting everything together, the optimization problem is:

$$
\begin{aligned}
\min_{\mathbf{v},\mathbf{m}} \quad & \sum_{p=1}^{P}\sum_{k=1}^{K} v_{p,k} & \\
\text{s.t.} \quad & \left| \sum_{\ell \in \mathcal{L}_{p,k}} \left(\sum_{t \in \mathcal{T}} m_{t,\ell} \right) - N_{p,k} \right| \leq v_{p,k}, & \forall p, k \\
& \sum_{t \in \mathcal{T}} m_{t,\ell} \leq 1, & \forall \ell \in \mathcal{L} \\
& \sum_{\ell \in \mathcal{L}} m_{t,\ell} = 1, & \forall t \in \mathcal{T} \\
& m_{t,\ell} \in \{0,1\}, & \forall t \in \mathcal{T}, \forall \ell \in \mathcal{L}
\end{aligned}
$$

Scalability

- The problem requires $T \times L + P \times K$ decision variables and a corresponding number of constraints.
- Since the matching variables grow quadratically with the number of units ($T \times L$), Gurobi v 7.5.1 with 32GB memory crashes

Linear-sized formulation

- Define a new binary variable for each treated unit:

$$z_\ell = \sum_{t \in \mathcal{T}} m_{t,\ell}, \quad \ell \in \mathcal{L}.$$

- Interpretation: $z_\ell = 1$ if treated unit ℓ is matched to a template unit, and 0 otherwise.
- Using z_ℓ, the last three constraints of the original MIP can be rewritten. The problem remains equivalent. The new problem:

$$\begin{aligned}
\min_{\mathbf{v},\mathbf{z}} \quad & \sum_{p=1}^{P} \sum_{k=1}^{K} v_{p,k} \\
\text{s.t.} \quad & \left| \sum_{\ell \in \mathcal{L}_{p,k}} z_\ell - N_{p,k} \right| \leq v_{p,k}, \qquad \forall p, k \\
& \sum_{\ell \in \mathcal{L}} z_\ell = T, \quad z_\ell \in \{0,1\}, \qquad \forall \ell \in \mathcal{L}
\end{aligned}$$

Theoretical Guarantees of the Linear Formulation

1. A Mixed-Integer Program (MIP) is considered **integral** if its Linear Programming (LP) relaxation naturally yields an integer-optimal solution.
2. The difference between the optimal objective value of the original integer problem and that of its LP relaxation is known as the **integrality gap** (or **LP gap**).

By shifting from a quadratic-size to a linear-size formulation, we do not sacrifice the tightness of the model. This is formalized in the following two propositions:

- **Proposition 4.1 (Equivalence of LP Gaps)** The linear-size MIP and the quadratic-size MIP share the exact same LP gap. The linear-size formulation is integral whenever the quadratic-size formulation is integral.
- **Proposition 4.2 (Integrality Guarantee for Two Covariates)** In scenarios where there are at most two covariates, the linear-size MIP is strictly integral.

Results: Computational Speed (Tables 1 and 2)

Tables 1 and 2 compare computational performance across different data sizes and levels of treatment granularity.

- Formulation 1 (quadratic size): Produced an out-of-memory error with 32GB of RAM in Gurobi v7.5.1.
- Formulation 2 (linear size): Successfully runs in R.

Results: Computational Speed (Tables 1 and 2)

Formulation 2 Results:

- **Table 1:** Template size varies from 1,000 to 10,000; total sample size from 70,000 to 700,000. The linear formulation remained scalable across all settings, with average computation time increasing from 0.2 minutes (70,000 observations) to 3.5 minutes (700,000 observations).
- **Table 2:** Template size fixed at 1,000; total sample size varies from 10,000 to 70,000. Different numbers of exposure levels (3, 5, 10) were also considered. Average computation time was under 0.3 minutes in all cases.

Together, these results demonstrate the substantial efficiency gains achieved by the linear-sized formulation.

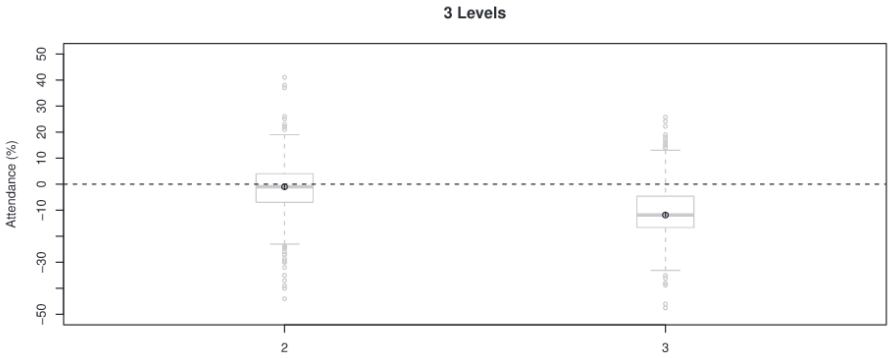
Results: Imbalance Measure (Table 3)

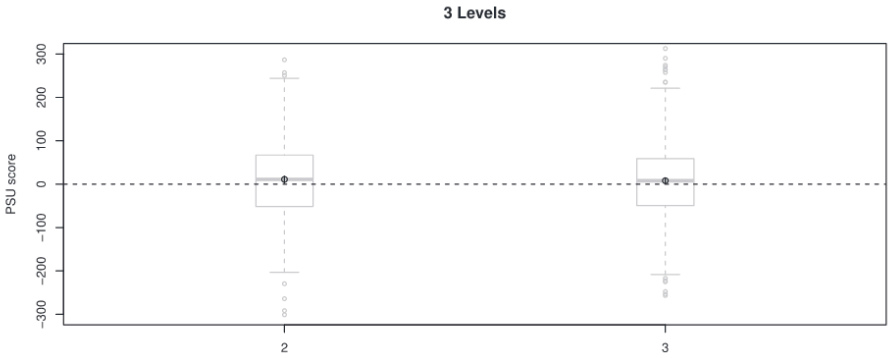
- Table 3 reports covariate imbalance under different numbers of treatment levels (3, 5, and 10).
- In all settings, the procedure achieved **exact matching**, meaning the objective value was zero in every case.

Results: Visualization of Causal Effects (Figure 2)

Overview:

- Target matched sample size: $\sim 1,000$ students.
- Procedure repeated with different random seeds to check stability.
- Outcomes examined:
  - Attendance significant effect observed.
  - Test scores no significant effect detected.


3 Levels
Attendance (%)
50
40
30
20
10
0
−10
−30
−50
2
3


3 Levels
PSU score
300
200
100
0
−100
−200
−300
2
3

Summary and Limitations

Summary:

- Introduced a scalable **cardinality matching** method using linear-sized MIP.
- Applied to Chilean earthquake data:
  - Found unexpected result on PSU test scores.
- Linear-sized formulation drastically improved **computational efficiency** without sacrificing exactness.

Limitations:

- No statistical guarantee (e.g. consistency, convergence rate, asymptotic distribution)...
- Only considers categorical variables